Brain Tumor MRI Classification

Comparing Classical ML, Custom CNN, and Transfer Learning

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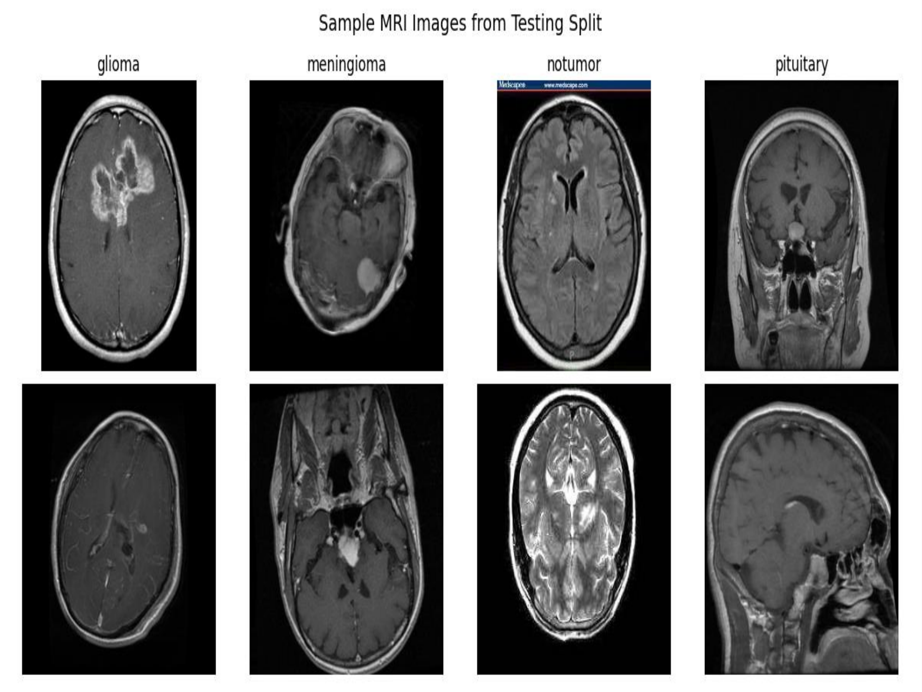
MTH/CSE 4224 — Intro to Machine Learning | Spring 2026

Problem & Dataset

4-class brain tumor classification from MRI scans:

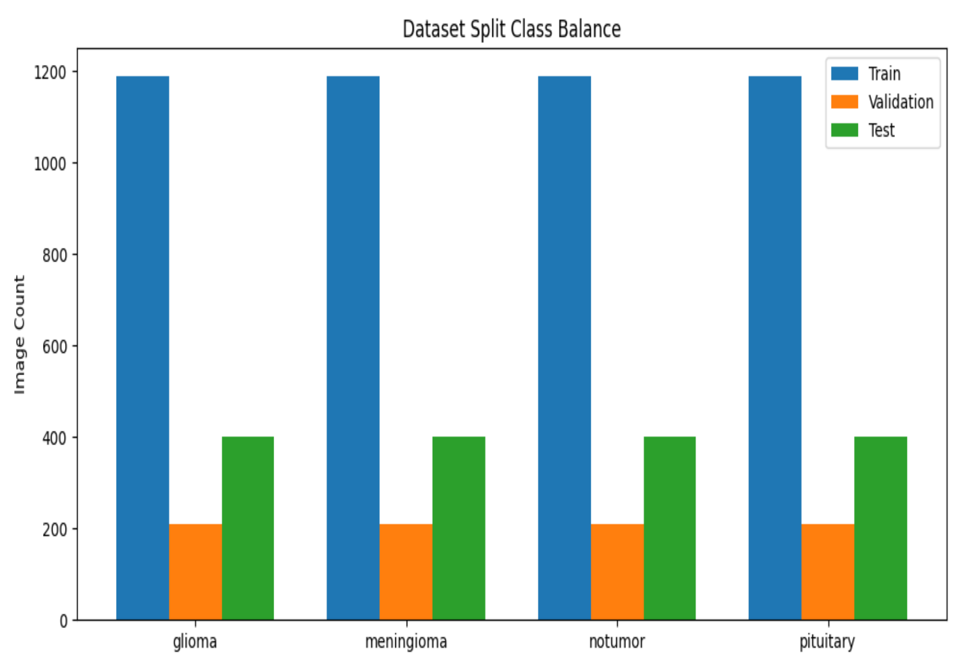
Glioma | Meningioma | Pituitary | No Tumor

- 7,200 MRI images from Kaggle dataset
- Balanced: 1,800 images per class
- Goal: compare baseline ML, custom CNN, and transfer learning



7,200 total images | **4** tumor classes | **224×224** pixel resolution

Data Split & Evaluation Policy



Split Strategy

- Training: 4,760 images (train) + 840 (validation)
- Testing: 1,600 images — held out, used once
- Perfectly balanced: 400 per class in test

Deviation from Proposal

- Originally planned 70/15/15 stratified split
- Instead kept Kaggle Testing/ as true held-out set
- **Sounder practice: prevents any data leakage**

Methods Overview

Baseline ML

Features: HOG + PCA

Dimensionality: 100-200

Models:

- Logistic Regression
- SVM (RBF)
- Random Forest

Custom CNN

Built from scratch

Conv + Pooling + FC layers

Batch size: 32

3 training epochs

Horizontal flip + rotation

Color jitter augmentation

Transfer Learning

ResNet50 backbone

ImageNet pretraining

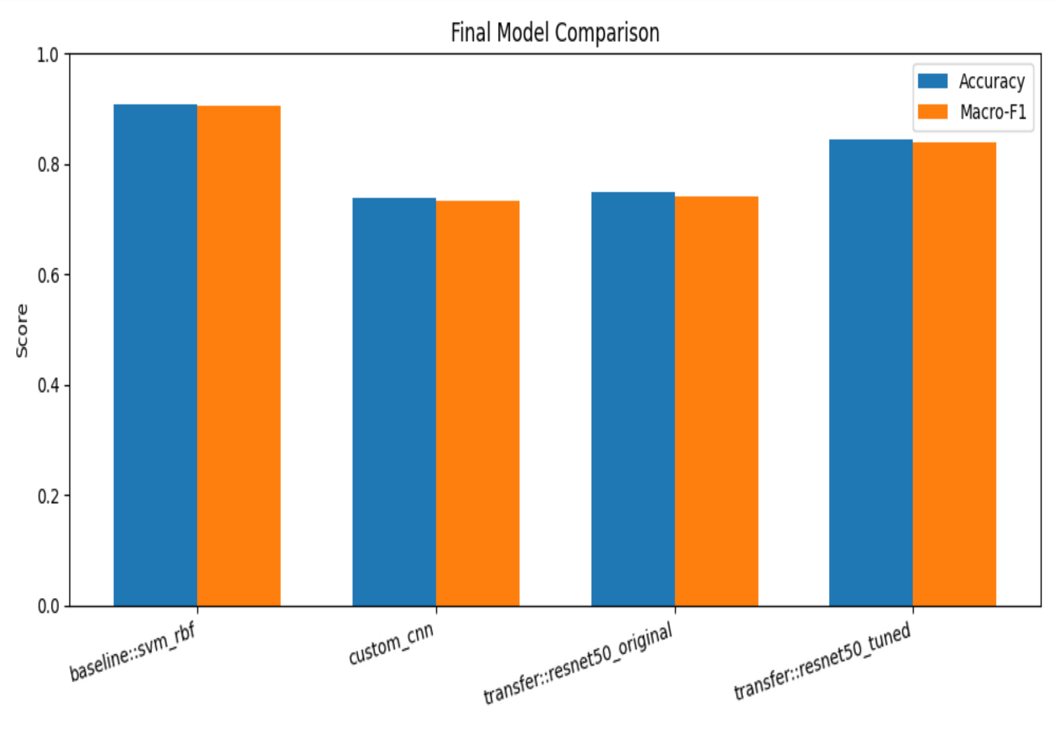
Two-stage tuning:

- Head warmup (frozen)
- Partial fine-tuning

Prevents catastrophic forgetting

Shared Metrics: Accuracy | Macro-Precision | Macro-Recall | Macro-F1

Final Results Comparison



SVM (RBF)

BEST OVERALL

Acc: 90.7%   F1: 90.5%

ResNet50 Tuned

Best Deep Model

Acc: 84.4%   F1: 84.0%

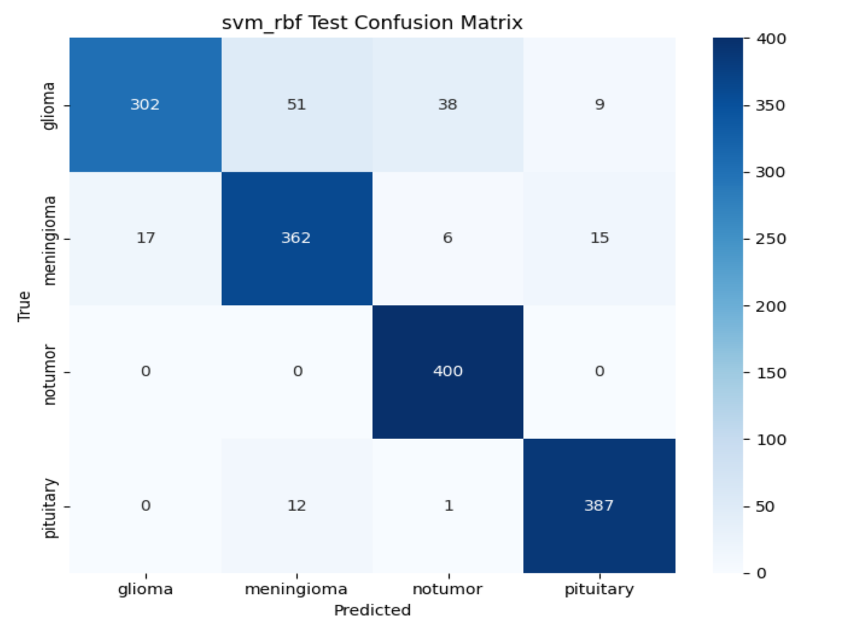
Random Forest

Acc: 86.8%   F1: 86.5%

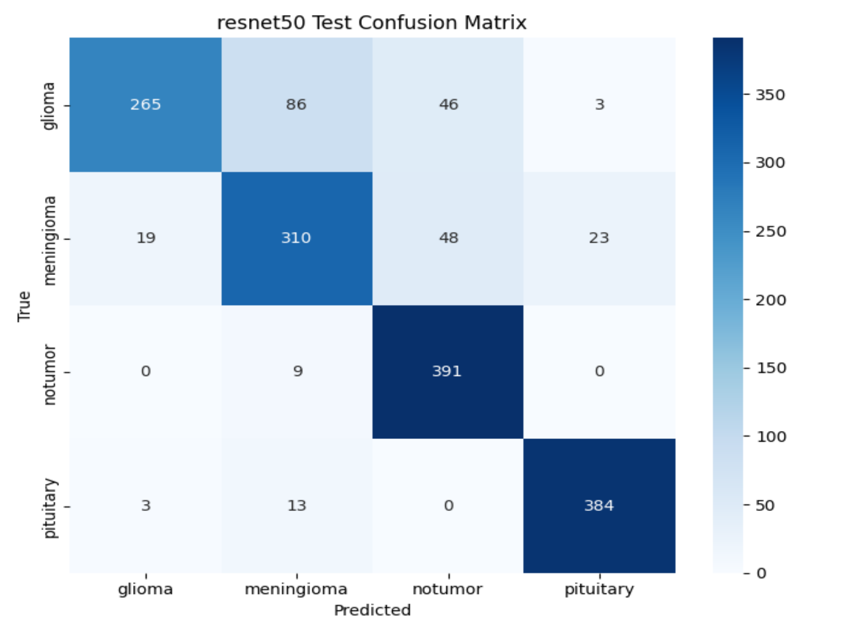
Custom CNN

Acc: 73.8%   F1: 73.3%

Error Analysis: Confusion Matrices



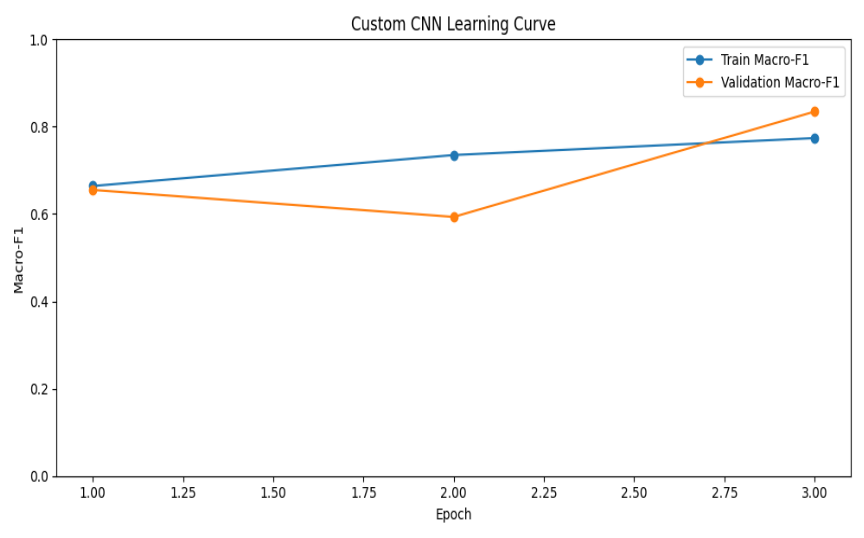
SVM (RBF) — Best Overall

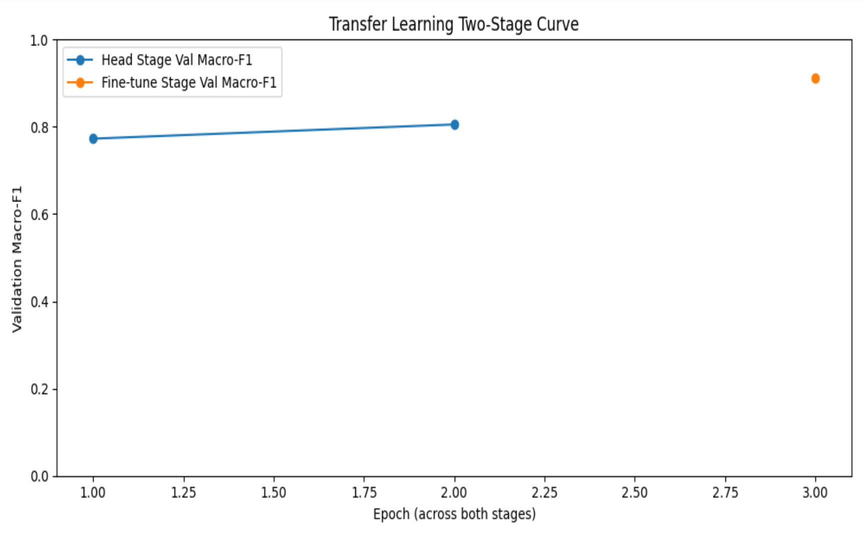


ResNet50 Tuned — Best Deep Model

Key Insight: "No tumor" is easiest class (SVM: 400/400 correct) | "Glioma" is hardest for all models

Training Dynamics





Custom CNN Learning Curve

Instability early, strong recovery by epoch 3

Transfer Learning Two-Stage Curve

Fine-tuning boosted val F1 from 0.81 to 0.91

Transfer Tuning Impact: **+9.5% accuracy** | **+10.0% macro-F1** (0.7488 → 0.8438 accuracy)

Key Methodological Decisions

1. **Held-out test set**

   Kept Kaggle Testing/ untouched for final evaluation only

2. **Macro-F1 as primary metric**

   Appropriate for multi-class medical classification

3. **HOG + PCA for baselines**

   Kept dimensionality manageable for classical ML

4. **Multiple baselines**

   LR, SVM, RF for thorough benchmarking against class methods

5. **Staged fine-tuning**

   Head warmup then partial unfreezing prevents catastrophic forgetting

Conclusions & Future Work

- SVM baseline achieves best performance: 90.7% accuracy, 90.5% macro-F1
- Transfer learning responds strongly to tuning (+9.5% accuracy improvement)
- Deep models did not surpass classical baseline on this dataset
- Disciplined evaluation: no test set leakage

Future Work

More training epochs | Larger augmentation pipeline | Ensemble methods | Additional architectures

Thank You

Questions?

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